

3.1 Poisson Manifolds and Symplectic Leaves

Definition 3.1.1 (Poisson manifold). A Poisson manifold is a pair $(P, \{\cdot, \cdot\})$, where P is a smooth manifold and $\{\cdot, \cdot\} : C^\infty(P, \mathbb{R}) \times C^\infty(P, \mathbb{R}) \rightarrow C^\infty(P, \mathbb{R})$ is $\mathbb{R}$ -bilinear, antisymmetric, satisfies the Jacobi identity, and the Leibniz rule

$$\{f, hk\} = \{f, h\}k + h\{f, k\}, \quad f, h, k \in C^\infty(P, \mathbb{R}).$$

Definition 3.1.2 (Hamiltonian vector field on a Poisson manifold). Let $(P, \{\cdot, \cdot\})$ be a Poisson manifold and $h \in C^\infty(P, \mathbb{R})$. The Hamiltonian vector field $\xi_h \in \mathfrak{X}(P)$ is defined by

$$\xi_h f := \{f, h\}, \quad f \in C^\infty(P, \mathbb{R}).$$

Definition 3.1.3 (Symplectic leaf). Let $(P, \{\cdot, \cdot\})$ be a Poisson manifold. A symplectic leaf $S_\alpha \subset P$ is a maximal connected immersed submanifold such that

$$T_\sigma S_\alpha = \{\xi_h(\sigma) \mid h \in C^\infty(P, \mathbb{R})\}, \quad \sigma \in S_\alpha,$$

and $\{\cdot, \cdot\}$ restricts to a nondegenerate bracket on S_α .

Lemma 3.1.4. Let $S_\alpha \subset P$ be a symplectic leaf with inclusion $\iota_\alpha : S_\alpha \hookrightarrow P$. If $f \in C^\infty(P, \mathbb{R})$ vanishes on S_α , then $\{f, h\}(\sigma) = 0$ for every $\sigma \in S_\alpha$, $h \in C^\infty(P, \mathbb{R})$.

Proof. ξ_h is tangent to S_α by Definition 3.1.3, and $f|_{S_\alpha} = 0$, so $\{f, h\}(\sigma) = \xi_h f(\sigma) = 0$. $\square$

Definition 3.1.5 (Poisson bracket on a symplectic leaf). For $f, h \in C^\infty(P, \mathbb{R})$, the Poisson bracket on S_α is

$$\{\iota_\alpha^* f, \iota_\alpha^* h\}_\alpha := \iota_\alpha^* \{f, h\}, \quad (3.1)$$

well defined by Lemma 3.1.4, independently of the extensions f, h .

Theorem 3.1 (Symplectic foliation). Every finite-dimensional Poisson manifold P decomposes as a disjoint union $P = \bigcup_\alpha \iota_\alpha(S_\alpha)$ of symplectic leaves as in Definition 3.1.3, each ι_α a Poisson immersion with respect to the bracket (3.1). The value $\{f, h\}(\sigma)$ depends only on the restrictions of f, h to the leaf through σ .

Proof. Existence of the leaves follows from the singular Frobenius theorem applied to the distribution spanned by the Hamiltonian vector fields, see [landsmanMathematicalTopicsBetween1998]. That ι_α is Poisson is (3.1). If $f'|_{S_\alpha} = f|_{S_\alpha}$, then $\{f, h\}(\sigma) = \{f', h\}(\sigma)$ for $\sigma \in S_\alpha$ by Lemma 3.1.4, and symmetrically for h . $\square$

3.2 Projective Hilbert Spaces as Symplectic Manifolds

Definition 3.2.1 (Compact operators). Let $\mathcal{H}$ be a Hilbert space as in Definition ???. $A \in \mathfrak{B}(\mathcal{H})$ as in Definition ?? is compact if it is a norm-limit of finite-rank operators. The compact operators form a closed two-sided ideal $\mathfrak{B}_0(\mathcal{H}) \subset \mathfrak{B}(\mathcal{H})$.

Definition 3.2.2 (Projective Hilbert space). Let $\mathcal{H}$ be a Hilbert space. On $\mathcal{H}^* := \mathcal{H} \setminus \{0\}$ define $\Psi \sim \Phi : \Leftrightarrow \exists z \in \mathbb{C} \setminus \{0\} : \Phi = z\Psi$. The projective Hilbert space is

$$\mathbb{P}\mathcal{H} := \mathcal{H}^* / \sim,$$

with $[\Psi]$ the equivalence class of $\Psi \in \mathcal{H}^*$.

Proposition 3.2.3. Let $S\mathcal{H} := \{\Psi \in \mathcal{H} \mid (\Psi, \Psi) = 1\}$. Restricting $\sim$ to $S\mathcal{H}$ gives $U(1)$ -orbits, and $[\Psi] \mapsto \tau(\Psi)$ is a bijection $\mathbb{P}\mathcal{H} \rightarrow S\mathcal{H}/U(1)$. Under it, $\mathbb{P}\mathcal{H}$ is identified with the pure state space of $\mathfrak{B}_0(\mathcal{H})$ as in Definition ??, and embeds strictly into the pure state space of $\mathfrak{B}(\mathcal{H})$ when $\dim \mathcal{H} = \infty$.

Proof. Every $[\Psi] \in \mathbb{P}\mathcal{H}$ has a unit representative, unique up to $z \in U(1)$, so $\tau : S\mathcal{H} \rightarrow \mathbb{P}\mathcal{H}$ is surjective with fibers the $U(1)$ -orbits. The identification with pure states of $\mathfrak{B}_0(\mathcal{H})$ and $\mathfrak{B}(\mathcal{H})$ follows from the correspondence between vector states and rank-one projections, [landsmanMathematicalTopicsBetween1998]. $\square$

Definition 3.2.4 (Local chart of $\mathbb{P}\mathcal{H}$). For $\psi \in \mathbb{P}\mathcal{H}$, $\Psi \in S\mathcal{H}$ a representative, and $N_\psi := \{\varphi \in \mathbb{P}\mathcal{H} \mid (\Psi, \Phi) \neq 0\}$ with Φ any representative of φ ,

$$F_\psi(\varphi) := \frac{\Phi}{(\Psi, \Phi)} - \Psi. \quad (3.2)$$

Proposition 3.2.5. F_ψ is well defined, $F_\psi(\psi) = 0$, F_ψ takes values in $\Psi^\perp$, and $F_\psi : N_\psi \rightarrow \Psi^\perp$ is a homeomorphism.

Proof. $\Phi/(\Psi, \Phi)$ is invariant under $\Phi \mapsto z\Phi$, $z \neq 0$, so F_ψ is well defined, and $F_\psi(\psi) = \Psi/(\Psi, \Psi) - \Psi = 0$. Also $(\Psi, F_\psi(\varphi)) = (\Psi, \Phi)/(\Psi, \Phi) - 1 = 0$. The map $\chi \mapsto \tau(\Psi + \chi)$, $\chi \in \Psi^\perp$, is a well-defined two-sided continuous inverse, since $\Psi + \chi \neq 0$ and both maps are built from continuous algebraic operations on the domain $(\Psi, \Phi) \neq 0$. $\square$

Varying ψ over a spanning set and fixing, for each, an isomorphism $\Psi^\perp \cong \mathcal{H}'$ with $\mathcal{H}'$ a reference Hilbert space of real dimension $\dim_{\mathbb{R}} \mathcal{H} - 2$, gives an atlas making $\mathbb{P}\mathcal{H}$ a Hilbert manifold modeled on $\mathcal{H}'$. We call the resulting topology the *manifold topology* of $\mathbb{P}\mathcal{H}$.

Proposition 3.2.6. The manifold topology on $\mathbb{P}\mathcal{H}$ coincides with the w^* -topologies relative to $\mathbb{P}\mathcal{H} \subset \mathfrak{B}_0(\mathcal{H})^*$ and $\mathbb{P}\mathcal{H} \subset \mathfrak{B}(\mathcal{H})^*$ as in Definition ??.

Proof. $\mathfrak{B}_0(\mathcal{H})$ is generated by rank-one projections, so its w^* -topology on $\mathbb{P}\mathcal{H}$ agrees with the weak topology on $\mathcal{H}$, which coincides with the strong topology on $S\mathcal{H}$, hence with the $\mathfrak{B}(\mathcal{H})^*$ w^* -topology. For $\varphi \in N_\psi$, $A \in \mathfrak{B}(\mathcal{H})$,

$$\frac{\varphi(A)}{\varphi([\Psi])} = (F_\psi(\varphi), AF_\psi(\varphi)) + (\Psi, AF_\psi(\varphi)) + (F_\psi(\varphi), A\Psi) + (\Psi, A\Psi). \quad (3.3)$$

Strong convergence $F_\psi(\varphi_n) \rightarrow F_\psi(\varphi)$ gives $\varphi_n(A) \rightarrow \varphi(A)$ for all A through (3.3), so the manifold topology is finer. Conversely $\varphi_n(A) \rightarrow \varphi(A)$ for all A forces weak convergence $F_\psi(\varphi_n) \rightarrow F_\psi(\varphi)$, and, taking $A = \mathbb{1}$, norm convergence, hence strong convergence. The two topologies coincide. $\square$

Corollary 3.2.7. The pure state space of $\mathfrak{B}_0(\mathcal{H})$ with the w^* -topology is homeomorphic to $\mathbb{P}\mathcal{H}$ with the manifold topology.

Proof. Combine Proposition 3.2.3 with Proposition 3.2.6. $\square$

Theorem 3.2 (Pure state decomposition). The pure state space $\mathcal{P}(\mathfrak{A})$ of a C^* -algebra $\mathfrak{A}$ as in Definition ?? is a disjoint union $\mathcal{P}(\mathfrak{A}) = \bigcup_\alpha \mathbb{P}\mathcal{H}_\alpha$, where $\mathcal{H}_\alpha$ is the Hilbert space of the irreducible GNS representation, Definition ??, of a state in $\mathbb{P}\mathcal{H}_\alpha$. States in the same $\mathbb{P}\mathcal{H}_\alpha$ are equivalent, states in different $\mathbb{P}\mathcal{H}_\alpha$ are inequivalent, and each inclusion $\mathbb{P}\mathcal{H}_\alpha \hookrightarrow \mathcal{P}(\mathfrak{A})$ is continuous.

Proof. Set-theoretic part: correspondence between pure states and irreducible GNS representations, together with Corollary 3.2.7. Continuity: Proposition 3.2.6. Full argument in [landsmanMathematicalTopicsBetween1 $\square$

Regard $\mathcal{H}$ as a real vector space, with $T_\Psi \mathcal{H} \cong \mathcal{H}$ through the directional derivative of Remark ??.

Definition 3.2.8. For $\Phi \in \mathcal{H}$, $V(\Phi) \in \mathfrak{X}(\mathcal{H})$ is given by

$$V(\Phi)_\Psi f := \left. \frac{d}{dt} \right|_{t=0} f(\Psi + t\Phi), \quad \Psi \in \mathcal{H}, f \in C^\infty(\mathcal{H}, \mathbb{R}).$$

If $V(\Phi)$ is tangent to $S\mathcal{H}$ at Ψ , its image under τ_* as in Definition ?? is $v(\Phi) \in T_\psi \mathbb{P}\mathcal{H}$.

Proposition 3.2.9. For $\psi = \tau(\Psi) \in \mathbb{P}\mathcal{H}$,

$$T_\psi \mathbb{P}\mathcal{H} = \{v(iA\Psi) \mid A \in \mathfrak{B}(\mathcal{H})_{\mathbb{R}}\}. \quad (3.4)$$

Proof. By Proposition 3.2.5, $T_\psi \mathbb{P}\mathcal{H} \cong \Psi^\perp$ through dF_ψ . Given $\Omega \in \Psi^\perp$, $A_\Omega := -i(|\Omega\rangle\langle\Psi| - |\Psi\rangle\langle\Omega|)$ is self-adjoint and $A_\Omega \Psi = -i\Omega$, so $iA_\Omega \Psi = \Omega$, giving $\Psi^\perp \subset \{iA\Psi\}$. Conversely, for self-adjoint A , $(\Psi, iA\Psi) = i(\Psi, A\Psi)$ equals $i(A\Psi, \Psi)$ by self-adjointness, forcing $\text{Re}(\Psi, iA\Psi) = 0$, so $iA\Psi \in \Psi^\perp$. Equation (3.4) follows. $\square$

Proposition 3.2.10. For $\hbar \in \mathbb{R} \setminus \{0\}$,

$$\omega(V(\Phi), V(\Omega)) := 2\hbar \operatorname{Im}(\Phi, \Omega) \quad (3.5)$$

is a symplectic form on $\mathcal{H}$ as in Definition ??.

Proof. Antisymmetry: $\overline{(\Phi, \Omega)} = (\Omega, \Phi)$. Closedness: constant coefficients. Nondegeneracy: $\omega(V(\Phi), V(i\Phi)) = 2\hbar \operatorname{Im}(i(\Phi, \Phi)) = 2\hbar \|\Phi\|^2$, zero only if $\Phi = 0$. $\square$

Definition 3.2.11. For $H \in \mathfrak{B}(\mathcal{H})_{\mathbb{R}}$, $\tilde{H} \in C^\infty(\mathcal{H}, \mathbb{R})$ is

$$\tilde{H}(\Psi) := (\Psi, H\Psi). \quad (3.6)$$

Proposition 3.2.12. The Hamiltonian vector field of $\tilde{H}$ with respect to ω , as in Definition ??, is

$$\xi_{\tilde{H}}(\Psi) = -V\left(\frac{i}{\hbar}H\Psi\right). \quad (3.7)$$

Proof. Self-adjointness of H gives

$$d\tilde{H}(\Psi)(V(\Phi)) = (\Phi, H\Psi) + (\Psi, H\Phi) = 2\operatorname{Re}(\Phi, H\Psi). \quad (3.8)$$

For $\xi(\Psi) := V(cH\Psi)$, $\omega(\xi(\Psi), V(\Phi)) = 2\hbar \operatorname{Im}(\bar{c}(H\Psi, \Phi))$. Taking $c = -i/\hbar$ gives $\bar{c} = i/\hbar$, $\operatorname{Im}(i(H\Psi, \Phi)) = \operatorname{Re}(H\Psi, \Phi)$, so $\omega(\xi(\Psi), V(\Phi)) = 2\operatorname{Re}(H\Psi, \Phi) = 2\operatorname{Re}(\Phi, H\Psi)$, matching (3.8). By $\iota_{\xi_{\tilde{H}}} \omega = d\tilde{H}$, $\xi_{\tilde{H}}(\Psi) = V(-iH\Psi/\hbar)$. $\square$

Proposition 3.2.13. The integral curve of $\xi_{\tilde{H}}$ through Ψ_0 solves the Schrödinger equation

$$i\hbar \dot{\Psi}(t) = H\Psi(t), \quad \Psi(t) = e^{-itH/\hbar} \Psi_0. \quad (3.9)$$

For $A, B \in \mathfrak{B}(\mathcal{H})_{\mathbb{R}}$, the Poisson bracket of Definition ?? satisfies

$$\{\tilde{A}, \tilde{B}\} = -\frac{i}{\hbar} \widetilde{[A, B]}. \quad (3.10)$$

Proof. (3.7) gives $\dot{\Psi} = -iH\Psi/\hbar$, i.e. (3.9). For (3.10), by (3.8) and (3.7),

$$\{\tilde{A}, \tilde{B}\}(\Psi) = d\tilde{A}(\Psi)(\xi_{\tilde{B}}(\Psi)) = 2\operatorname{Re}\left(-\frac{i}{\hbar}B\Psi, A\Psi\right) = -\frac{2}{\hbar} \operatorname{Im}(B\Psi, A\Psi).$$

Since $(\Psi, [A, B]\Psi) = (A\Psi, B\Psi) - (B\Psi, A\Psi) = -2i \operatorname{Im}(B\Psi, A\Psi)$, this is $\frac{i}{\hbar}(\Psi, [A, B]\Psi) \cdot (-1) = -\frac{i}{\hbar}(\Psi, [A, B]\Psi)$. $\square$

Remark 3.2.14. Sign opposite to [landsmanMathematicalTopicsBetween1998], due to the orientation $\iota_{X_H} \omega = dH$ of Definition ??. Check against the canonical commutation relation convention of the Weyl algebra chapter.

Remark 3.2.15. For U unitary as in Definition ??, $U^* \tilde{A} = \widetilde{U^{-1}AU}$, so U acts on $(\mathcal{H}, \omega)$ as a symplectic map, Definition ??.

Proposition 3.2.16. $U(1)$ acts freely on $\mathcal{H}^*$, so $\mathcal{H}^*/U(1)$ is a Poisson manifold as in Definition 3.1.1, with symplectic leaves $S_r := \mathcal{H}_r/U(1)$, $\mathcal{H}_r := \{\Psi \mid (\Psi, \Psi) = r^2\}$, as in Theorem 3.1. $\mathbb{P}\mathcal{H} \cong S_1$, symplectic, with

$$\omega_\psi(v(iA\Psi), v(iB\Psi)) = -i\hbar \widetilde{[A, B]}(\psi). \quad (3.11)$$

Proof. $z\Psi = \Psi$, $\Psi \neq 0$ forces $z = 1$, so the action is free. $\Psi \mapsto (\Psi, \Psi)$ descends and its level sets are the $\mathcal{H}_r$. The Poisson tensor of $(\mathcal{H}, \omega)$ descends since $U(1)$ acts by symplectic maps, Remark 3.2.15, and its symplectic leaves are the S_r by Theorem 3.1. $S\mathcal{H}/U(1) = S_1$ by $(\Psi, \Psi) = 1$. Explicit form (3.11), [landsmanMathematicalTopicsBetween1998]. $\square$

Definition 3.2.17. For $H \in \mathfrak{B}(\mathcal{H})_{\mathbb{R}}$, $\hat{H} \in C^\infty(\mathbb{P}\mathcal{H}, \mathbb{R})$ is $\hat{H}(\tau(\Psi)) := \tilde{H}(\Psi)$, $\Psi \in S\mathcal{H}$.

Remark 3.2.18. Well defined since $\tilde{H}(z\Psi) = \tilde{H}(\Psi)$ for $z \in U(1)$. $\|H\| = \|\hat{H}\|_\infty$, the Gelfand transform of H , Theorem ??, $\xi_{\hat{H}}(\tau(\Psi)) = -v(iH\Psi/\hbar)$, $\{\hat{A}, \hat{B}\} = -\frac{i}{\hbar}[A, B]$, every unitary acts as a symplectic map on $\mathbb{P}\mathcal{H}$. For $\mathcal{H} = \mathbb{C}^N$, (3.11) is $\hbar$ times the Fubini-Study form.

Proposition 3.2.19. $\psi(t) := \tau(\Psi(t))$, with $\Psi(t)$ as in (3.9), is the complete flow of $\xi_{\hat{H}}$:

$$\frac{d\psi(t)}{dt} = \xi_{\hat{H}}(\psi(t)), \quad \psi(t) = e^{-it\hat{H}/\hbar}\psi(0).$$

Proof. τ is a submersion on $S\mathcal{H}$, so $\dot{\psi}(t) = v(\dot{\Psi}(t)) = \xi_{\hat{H}}(\psi(t))$. Completeness: $\Psi(t) = e^{-itH/\hbar}\Psi(0)$ is defined for all $t \in \mathbb{R}$. $\square$

Proposition 3.2.20. $\Psi \in S\mathcal{H}$ is an eigenvector of $H \in \mathfrak{B}(\mathcal{H})_{\mathbb{R}}$ iff $\psi = \tau(\Psi)$ is a critical point of $\hat{H}$, $d\hat{H}(\psi) = 0$, with eigenvalue $\hat{H}(\psi)$.

Proof. By Proposition 3.2.9, $d\hat{H}(\psi) = 0$ iff $\text{Re}(\Psi, (HA - AH)\Psi) = 0$ for all self-adjoint A , that is, $(\Psi, AH\Psi) \in \mathbb{R}$ for all such A . By the surjectivity in the proof of Proposition 3.2.9, every $\Phi \in \Psi^\perp$ equals $A\Psi$ for some self-adjoint A , and every $i\Phi$ equals $A'\Psi$ for some self-adjoint A' , forcing $(\Psi, H\Phi) \in \mathbb{R}$ and $i(\Psi, H\Phi) \in \mathbb{R}$, hence $(\Psi, H\Phi) = 0$ for all $\Phi \in \Psi^\perp$. This means $H\Psi \in (\Psi^\perp)^\perp = \mathbb{C}\Psi$, that is, Ψ is an eigenvector with eigenvalue $(\Psi, H\Psi) = \hat{H}(\psi)$. $\square$

3.3 Representations of Poisson Algebras

Definition 3.3.1 (Representation of a Poisson algebra). Let $(\mathfrak{A}_{\mathbb{R}}, \circ, \{\cdot, \cdot\})$ be a Poisson algebra with Jordan product $\circ$, Definition ??, A representation is a linear map $\pi : \mathfrak{A}_{\mathbb{R}} \rightarrow C^\infty(S, \mathbb{R})$, S symplectic as in Definition ??, such that

$$\pi(f \circ h) = \pi(f)\pi(h), \quad (3.12)$$

$$\{\pi(f), \pi(h)\}_S = \pi(\{f, h\}), \quad (3.13)$$

for $f, h \in \mathfrak{A}_{\mathbb{R}}$, and such that π preserves completeness: the flow of $\xi_{\pi(h)}$ is defined for all times whenever h is complete in $\mathfrak{A}_{\mathbb{R}}$.

Definition 3.3.2 (Equivalence of representations). $\pi_1 : \mathfrak{A}_{\mathbb{R}} \rightarrow C^\infty(S_1, \mathbb{R})$ and $\pi_2 : \mathfrak{A}_{\mathbb{R}} \rightarrow C^\infty(S_2, \mathbb{R})$ are equivalent if there is a symplectomorphism $J : S_1 \rightarrow S_2$ with $J^*\pi_2(f) = \pi_1(f)$ for all $f \in \mathfrak{A}_{\mathbb{R}}$.

Definition 3.3.3 (Poisson space). A Poisson space P is a Hausdorff space with a linear subspace $\mathfrak{A}_{\mathbb{R}} \subset C(P, \mathbb{R})$ and symplectic manifolds $\{S_\alpha\}$, Definition ??, the symplectic leaves of P , with continuous injections $\iota_\alpha : S_\alpha \hookrightarrow P$, such that:

- (1) $P = \bigcup_\alpha \iota_\alpha(S_\alpha)$, disjoint union;
- (2) $\mathfrak{A}_{\mathbb{R}}$ separates points of P ;
- (3) $\mathfrak{A}_{\mathbb{R}} \subset C_L^\infty(P, \mathbb{R}) := \{f \in C(P, \mathbb{R}) \mid \iota_\alpha^* f \in C^\infty(S_\alpha, \mathbb{R}) \forall \alpha\}$;
- (4) $\mathfrak{A}_{\mathbb{R}}$ is closed under

$$\{f, h\}(\iota_\alpha(\sigma)) := \{\iota_\alpha^* f, \iota_\alpha^* h\}_\alpha(\sigma), \quad \sigma \in S_\alpha, \quad (3.14)$$

with $\{\cdot, \cdot\}_\alpha$ as in Definition 3.1.5.